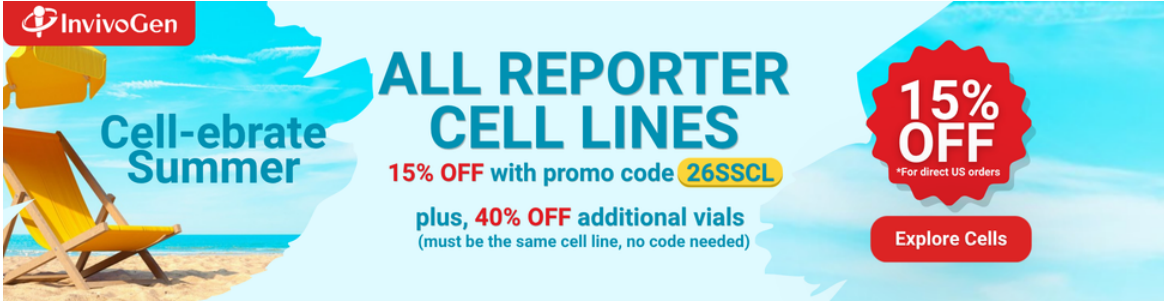


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Recombinant SpTrf proteins illustrate putative functions of native SpTrf proteins: from immune response to resolution in sea urchins

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Abstract

The complex and sophisticated sea urchin innate immune system relies on conserved immune receptor and effector molecules plus the euechinoid-specific Transformer family. Native SpTransformer (natSpTrf) proteins in *Strongylocentrotus purpuratus* respond to immune challenge by augmenting phagocytosis of microbes. These proteins show significant sequence diversity both within and among sea urchins with the predicted benefit of host protection through the recognition of a broad range of pathogens. Prior reports show that 6 recombinant (r)SpTrf proteins from insect cells induce phagocytosis when cross-linked to inert beads. Soluble rSpTrf proteins bind phagocytes, which respond by downregulating expression of the *SpTrf* gene family and *SpIL17-9*. Here we extend this analysis to determine the rSpTrf protein binding targets and find that they fail to bind pathogen-associated molecular patterns (PAMPs) and bacteria. Similarly, Ni-isolated natSpTrf proteins also fail to bind the same PAMPs. The insect cells add posttranslational modifications to the rSpTrf proteins that impart protein stability and impact predicted structure. Together with prior results, our findings suggest a working hypothesis for the natSpTrf proteins in vivo in which disordered natSpTrf proteins bind to pathogens, transform to secondary structure, and augment phagocytosis leading to pathogen clearance. Upon pathogen clearance, unbound natSpTrf proteins transform, multimerize, and bind to phagocytes. The cellular response adjusts expression of the *SpTrf* genes and *SpIL17-9* to the threat level. This hypothesis suggests that the natSpTrf proteins participate in immune responses to foreign challenge but also function in immune resolution after pathogen clearance.

Keywords echinoderm, intrinsically disordered proteins, pathogen binding, *Strongylocentrotus purpuratus*

Introduction

The sophisticated innate immune system in sea urchins relies on unique effector molecules, a subset of innate immune proteins conserved in vertebrates, and expansions of several homologous gene families including Toll-like receptors and nucleotide-binding oligomerization domain-like receptors, which recognize pathogen-associated molecular patterns (PAMPs)^{1,2} (reviewed in Buckley and Rast^{3,4}). The expanded sequence diversity of immune proteins results in broad binding and effector functions that leads to effective host protection (eg Chiaramonte et al.,⁵ Buckley and Dooley⁶) and likely controls the host-associated microbiomes.^{7–10} One of the expanded gene families is the *SpTransformer* (*SpTrf*) genes in the purple sea urchin from the Pacific coast of North America, *Strongylocentrotus purpuratus* (reviewed in Smith and Lun¹¹). The *SpTrf* gene family was initially

identified based on significantly elevated expression in response to challenge with marine bacteria,^{12,13} which was verified by gene expression in response to several PAMPs.^{14–16} *Trf* gene families are present in other euechinoids such as those in the eastern Pacific purple sea urchin, *Heliocidaris erythrogramma*, and the Mediterranean purple sea urchin, *Paracentrotus lividus*.^{17,18} Preliminary genome searches also identify *Trf* sequences in the painted sea urchin, *Lytechinus pictus* (eg accession number XM_064099691 [note, the encoded LpTrf protein is incorrectly annotated as a Ser/Thr protease]). The *SpTrf* gene family in the *S. purpuratus* genome is composed of 9 small genes, each with 2 exons and with dissimilar alleles that have also been evaluated from genomic DNA inserts in bacterial artificial chromosomes (BACs).^{19–21} However, when sea urchin genomic DNA is evaluated for conserved *SpTrf* sequences using several molecular methods and statistical estimates, many more gene sequences are

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predicted than those in the BAC inserts.^{16,22} The sequence diversity of the *SpTrf* gene family is based on multiple single-nucleotide polymorphisms plus blocks of similar sequence termed “elements” that are present in mosaic-like patterns in the coding regions (Fig. 1).^{14–16,23} In many cases, the elements correspond to repeats, along with shared elements among genes and repeats that surround the genes. This structure has been proposed to lead to increased sequence diversity through gene recombination, duplication, and deletion.^{21,22} In addition to the sequence diversity among genes within individual sea urchins, they also show significant diversity among animals,²⁴ which is amplified by transcript editing.²⁵ These characteristics of the *SpTrf* gene family are consistent with an “arms race” between host and pathogens, leading to diversity in sequences and function.

The SpTrf proteins all have the same general structure of a leader followed by a glycine (Gly)-rich region, a multimerization region, a histidine (His)-rich region, and a C-terminal region (Fig. S1).^{15,26} The structure of the mature SpTrf proteins, excluding the leader, is predicted to be intrinsically disordered based on in silico analysis and as confirmed by circular dichroism (CD) of one recombinant version.^{26,27} The native (nat)SpTrf proteins are expressed by the phagocyte class of coelomocytes and show elevated expression in subsets of the small and polygonal phagocytes in response to immune challenge.^{28–30} Most of the natSpTrf proteins multimerize irreversibly upon collection from the coelomic fluid (CF), which deactivates their functions.^{26,28,31} Previous reports have shown that the natSpTrf proteins bind to a wide range of bacteria³² (see the supplementary file published with Lun et al.²⁶), including the Gram-negative marine microbe *Vibrio diazotrophicus*, on which they function as opsonins to augment phagocytosis.²⁹ The His-rich region of SpTrf proteins has been used for isolation from the CF by immobilized metal (nickel) affinity chromatography (IMAC).^{29,33} However, many distinct natSpTrf proteins are isolated with this approach (Fig. S2), which does not enable investigations of individual natSpTrf proteins. Consequently, functional studies were initiated with an rSpTrf protein with an E1 element pattern (Fig. 1) expressed and isolated from *Escherichia coli* (rSpTrf-E1-Ec).²⁶ rSpTrf-E1-Ec binds lipopolysaccharide (LPS), flagellin, phosphatidic acid (PA; a phospholipid), *V. diazotrophicus*, *Saccharomyces cerevisiae*, and β -1,3-glucan; however, it does not bind peptidoglycan or

Bacillus spp.^{26,27,31} Upon binding targets such as LPS or PA, rSpTrf-E1-Ec undergoes a structural transformation from disordered to α -helical²⁷ or it multimerizes into aggregates leading to loss of function.³¹ Proteins with multiple binding targets are consistent with intrinsically disordered structures (reviewed in Uversky³⁴ and Holehouse et al.³⁵) that adopt different structures to enable binding to distinct targets.³⁶ Based on this unexpected characteristic of rSpTrf-E1-Ec and the hypothesis that individual natSpTrf proteins undergo structural transformations, the proteins were renamed from the original of Sp185/333 to SpTransformer with a concurrent renaming of the genes and transcripts²⁷ (reviewed in Smith and Lun¹¹). Although rSpTrf-E1-Ec has been described as multitasking, or has the ability to bind to various types of targets, it only shows half of the expected functions of an immune response protein because it does not augment phagocytosis and therefore is unlikely to bind to phagocytes.²⁹

rSpTrf-E1-Ec from *E. coli* does not carry posttranslational modifications (PTMs), which is different from the natSpTrf proteins and may impact function. Based on the unexpected failure to augment phagocytosis, and because 5 other versions of rSpTrf proteins are toxic to *E. coli*,²⁶ 6 different rSpTrf proteins have been expressed successfully in insect cells.³⁷ Prior results show that these proteins induce phagocytosis by a subset of phagocytes when they are cross-linked to inert beads, which is mediated through direct and specific interactions between the rSpTrf proteins and shared binding sites on phagocyte cell surfaces.^{30,37} The phagocytes respond to a subset of the bound rSpTrf proteins by modulating the expression of the *SpTrf* gene family and the proinflammatory *SpIL17-9* gene.³⁸ Although the rSpTrf proteins from insect cells bind to phagocytes, whether they bind to a range of foreign targets has not been explored. Here we address this aspect of the functions of 6 rSpTrf proteins from insect cells (Fig. 1, Fig. S1) and build on prior results. The findings presented here indicate that the rSpTrf proteins have PTMs added by the insect cells and they do not bind PAMPs or microbes. Interestingly, this matches results for IMAC-isolated natSpTrf proteins that also fail to bind PAMPs. The 3 anti-natSpTrf antibodies raised against 3 relatively conserved peptides in the SpTrf sequences (Fig. S1)²⁸ show variable binding to each of the rSpTrf proteins, which is in contrast with binding to natSpTrf proteins. Structure predictions suggest that the rSpTrf

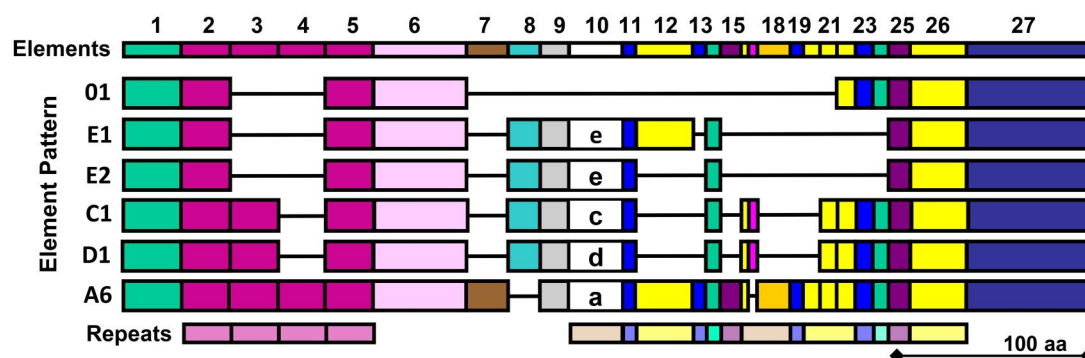


Figure 1. The cartoon alignment shows the sequence diversity of the rSpTrf proteins based on the variety of element patterns. The rSpTrf proteins that are investigated in this study are shown in the alignment. Protein names are indicated to the left and reflect the differences in the element patterns. The maximum number of elements from all known SpTrf sequences are shown at the top.²⁵ Tandem and interspersed repeats are shown at the bottom. Repeats that correspond to elements are color-coded similarly. This figure is modified from Buckley et al.²⁵

proteins are disordered; however, unlike rSpTrf-E1-Ec, they remain disordered and do not transform in the presence of sodium dodecyl sulfate (SDS). The PTMs added by insect cells may constrain their conformation so that they do not bind foreign targets, and may be the basis for binding to phagocytes.³⁰ Overall, functions for rSpTrf proteins from insect cells and those for rSpTrf-E1-Ec^{26,29} do not align.

The integration of the results for rSpTrf proteins from *E. coli* versus insect cells that we report here provides the basis for a hypothesis of the immunological functions of the natSpTrf proteins in sea urchins. Upon pathogen detection by phagocytes, the *SpTrf* gene family is upregulated and the natSpTrf proteins are secreted into the CF. The proteins bind to pathogens, transform, multimerize, bind to phagocytes, and augment phagocytosis. Pathogen clearance leads to an excess of unbound natSpTrf proteins in the CF that multimerize and bind to phagocytes. The cells respond by downregulating *SpTrf* gene family expression that is involved in the resolution of the immune response.

Materials and methods

Expression and isolation of rSpTrf proteins

A set of rSpTrf proteins (Table S1) were expressed and secreted by Sf9 insect cells and isolated by nickel affinity chromatography or IMAC as described previously.³⁷ In brief, coding regions minus the leader were cloned into the pMIB-V5-His A expression vector (Thermo Fisher; item V803001) by Gibson Assembly or by classical ligation into the *Hind*III and *Xho*I restriction sites and transfected into Sf9 insect cells. The rSpTrf proteins were isolated from concentrated culture media by IMAC and evaluated by spectrophotometry using the Scopes method³⁹ and by western blots (see below) with a rabbit anti-V5 antibody linked to horseradish peroxidase (α -V5-HRP, Thermo Fisher; item 50273009) as reported previously.³⁷

Western blots

rSpTrf proteins (1 to 2 μ g) were separated by 8% to 12% SDS-PAGE with broad range standards (Bio-Rad Laboratories, item 1610373) and transferred to polyvinylidene fluoride (PVDF) filters (Immobilon-P^{seq}, EMD Millipore, item ISEQ85R) for 10 min in a Trans-blot Turbo Transfer System (Bio-Rad Laboratories, item 1704150). Filters were rinsed in Tris-NaCl-Tween (TNT; 25 mM Tris pH 7.4, 0.5 M NaCl, 0.1% Tween 20) and blocked in blotto (TNT plus 5% milk) with rocking for 1.5 h at room temperature (rt) or overnight at 4°C. rSpTrf proteins were detected with α -V5-HRP (1500 $\times$ to 3000 $\times$ dilution in blotto). natSpTrf proteins on western blots were detected with 3 primary rabbit antibodies, anti-natSpTrf-66 (α -66), α -68, and α -71 (3000 $\times$ dilution each in blotto) that recognized different regions of the SpTrf proteins²⁸ (Fig. S1; see also the supplementary file in Crow et al.³⁷) followed by the secondary antibody (goat-anti-rabbit-Ig-HRP [G α Rlg-HRP], 3000 $\times$ dilution in blotto; Abcam; item ab6721) with rocking for 1 h at rt as reported.²⁸ Filters were washed twice in TNT and twice in Tris-NaCl without Tween 20 (TN), incubated in enhanced chemiluminescence (ECL) solution (Super Signal

West Pico PLUS, Thermo Scientific; item 34580), and evaluated for chemiluminescence in a ChemiDoc Touch Imaging System (Bio-Rad Laboratories, item 17001401JA).

Sea urchins

Adult sea urchins, *Strongylocentrotus purpuratus*, were purchased from the Southern California Sea Urchin Company (Corona del Mar, CA, USA) after collection from the coast of southern California near San Diego. Sea urchins were fed rehydrated kelp (*Saccharina angustata*; Wel-Pac Dashi Kombu Dried Seaweed, Amazon) and the seawater aquaria were maintained as described previously.^{7,40,41}

Isolation of natSpTrf proteins by Ni affinity

natSpTrf proteins were isolated by IMAC as described previously.³³ CF was collected from 12 to 15 sea urchins in calcium- and magnesium-free seawater with 30 mM EDTA and 20 mM imidazole pH 7.4 (CMFSW-EI; 460 mM NaCl, 10.7 mM KCl, 7.04 mM Na₂SO₄, 2.38 mM NaHCO₃)¹⁶ and the cells were pelleted. The supernatant, or cell-free CF (cfCF), was stored at -80°C . cfCF (35 mL) was incubated with Ni-NTA agarose beads (1.75 mL; Invitrogen, item R90115) at rt for 1 h with rotation. Beads were pelleted at $800 \times g$, loaded into a 5-mL polypropylene column (Thermo Fisher), and washed 3 times with 7 mL binding-wash buffer (10 mM NaCl, 50 mM NaPO₄ pH 8, 40 mM imidazole). Bound proteins were eluted with 1 mL elution buffer (10 mM NaCl, 50 mM NaPO₄ pH 8, 300 mM imidazole), which was repeated 7 times. The IMAC-isolated natSpTrf proteins and unbound proteins in the column flowthrough were evaluated by western blot with α -66, α -68, and α -71, followed by G α Rlg-HRP as described above. Eluted proteins were combined, loaded into a Slide-a-Lyzer mini dialysis unit (10,000 molecular weight cutoff; Thermo Fisher, item 69570), and concentrated against polyethylene glycol 8000 (Thermo Scientific, item 043443.A9) for 1 to 2 h at 4°C followed by dialysis against PBS (137 mM NaCl, 2.7 mM KCl, 7.7 mM Na₂HPO₄, 2.3 mM NaH₂PO₄, pH 7.4) overnight at 4°C. Protein concentration was evaluated at OD₂₀₅ according to the Scopes method,³⁹ and aliquots were stored at -80°C .

ELISA

rSpTrf protein incubation with flagellin or LPS

Flagellin (100 ng from *Salmonella typhimurium*; Invivogen, item tlrI-bsfla) or LPS (200 pg from *E. coli* O55:B5; Invivogen, item tlrI-b5lps) in 100 μ L PBS were loaded into wells of a 2 HB 96-well plate (Fisher Scientific, item 3455) and incubated overnight with rocking at 4°C. Control wells were incubated with PBS that omitted flagellin or LPS. The wells were washed once with 300 μ L PBS with 0.5% Tween (PBST) and blocked with 300 μ L PBS with 5% milk (PBSM) with rocking for 1 h at rt. The block was removed and 20 nM of each rSpTrf protein in 100 μ L PBSM was added to the wells and incubated with rocking for 1 h at rt. The wells were washed 3 times with 300 μ L PBST for 3 min each wash. Wells were incubated with α -V5-HRP (8000 $\times$ dilution in PBSM) with rocking for 1 h at rt. The wells were washed as above and incubated with 3,3',5,5'-tetramethylbenzidine (TMB, 100 μ L;

Thermo Fisher, item 34029) in the dark for 20 to 30 min. Stop solution (100 μ L, 1 N HCl) was added and the plate was read immediately at OD₄₅₀ in a Synergy HTX plate reader (BioTek, item 31081).

rSpTrf incubation with bacteria

Vibrio diazotrophicus^{42,43} was inoculated into 4 mL marine broth 2216 (Difco Laboratories) and *Bacillus cereus* was inoculated into 4 mL Luria Bertani broth. Both were incubated with rotation at rt for 18 h.

In-cell ELISA was used to evaluate rSpTrf binding to bacteria according to published methods for intact *Vibrio cholerae*.^{44,45} In brief, wells in 2HB plates were coated with 50 μ L of 0.001% poly-L-lysine (PLL) in 50 mM carbonate buffer (pH 9.2 to 9.6; 4 mM Na₂CO₃, 46 mM NaHCO₃) and incubated at 4 °C overnight. *Vibrio* and *Bacillus* were diluted with PBS and centrifuged at 15,300 $\times$ g for 5 min at 16 °C to 18 °C. The PLL was removed from the wells, the bacterial pellets were resuspended in PBS, and 50 μ L (the equivalent of 10 μ L of culture) was distributed into individual wells. Control wells received PBS without bacteria. The plate was centrifuged at 1,000 $\times$ g for 20 min at 16 °C to 18 °C, and the bacteria were fixed with 2% paraformaldehyde (50 μ L; EM Sciences, item 15710) at rt for 15 min. The fixative and unbound bacteria were removed, and 0.1 M glycine (100 μ L) was added per well and incubated at rt for 30 min. The wells were washed once with 200 to 300 μ L PBST for 3 min, and blocking solution (100 μ L; 3% BSA, 3% normal rabbit serum in PBST) was added and incubated with rocking at 4 °C overnight or at rt for at least 2 h. The wells were washed as above and rSpTrf proteins or recombinant frog colony-stimulating factor 1 with a V5 tag (rCSF-1-V5)^{46,47} (30 nM proteins in 100 μ L block) were added to wells and incubated with rocking for 1 h at rt. The wells were washed and α -V5-HRP (5000 $\times$ dilution in 100 μ L block) was added to the wells and incubated with rocking for 1 h at rt. The wells were washed, TMB (100 μ L) was added to each well, and the color reaction was developed for 20 min at rt in the dark. Stop solution (100 μ L, 1 N HCl) was added and the color reaction was evaluated at OD₄₅₀. Background readouts for wells with PBS only were subtracted from readouts from wells with bacteria.

rSpTrf protein incubation with rabbit anti-natSpTrf antibodies

Each rSpTrf protein in 100 μ L PBS was loaded into wells of 2HB plates in a concentration series (0, 0.1, 0.3, 1.0, 2.0 nM). natSpTrf proteins isolated by IMAC from cfCF in 100 μ L PBS were also loaded into wells in a concentration series (0, 0.1, 2.5, 12.5, 25, 50 ng). The plate was incubated with rocking overnight at 4 °C. The wells were washed 3 times with 300 μ L PBST for 3 min each. Wells were blocked with 300 μ L PBSM with rocking for 1 h at rt. The wells were washed twice with 300 μ L PBST for 3 min each. Primary antibodies, α -66, α -68 (3000 $\times$ dilution), or α -71 (30,000 $\times$ dilution) in 100 μ L of PBSM were added to the wells and incubated with rocking for 1 h 20 min at rt. The wells were washed 3 times with PBST and G α Rlg-HRP (15,000 $\times$ dilution in 100 μ L PBSM) was added to the wells and incubated with rocking for 1 h at rt. The wells were washed and TMB (100 μ L) was added to the wells and incubated in the dark for 10 min at rt. The stop solution (100 μ L, 1 N HCl) was added and the color reaction was evaluated at OD₄₅₀.

Deglycosylated rSpTrf protein incubation with 3 rabbit anti-natSpTrf antibodies

The rSpTrf proteins rSpTrf-E1, -E2-3, and -E2-4 were deglycosylated with PNGaseF (New England Biolabs, item P0704S) according to the supplier recommendation and as described previously.³⁰ Controls included proteins incubated in the reaction buffer that omitted PNGaseF, and proteins that were untreated. Afterward, proteins (2 nM in 100 μ L PBS) were loaded into wells of a 2HB plate and incubated with rocking overnight at 4 °C. Binding by the 3 antibodies, α -66, α -68 (2000 $\times$ dilution in PBSM), and α -71 (40,000 $\times$ dilution in PBSM) followed by G α Rlg-HRP was evaluated by ELISA as described above.

Glycosylated and deglycosylated rSpTrf-E1 incubation with flagellin

Flagellin (0, 2, 4 μ g) in 100 μ L PBS was incubated in wells of a 2HB plate overnight with rocking at 4 °C as described above. Wells were washed once with PBST, twice with coelomocyte culture medium (CCM; 0.5 M NaCl, 5 mM MgCl₂, 1 mM EGTA, and 20 mM HEPES, pH 7.2),⁴⁸ and blocked with CCM with 5% milk (CCM-M) with rocking for 1 h at rt. Wells were washed with CCM, and glycosylated and deglycosylated rSpTrf-E1 (2 μ g; 0.55 μ M) in 100 μ L CCM-M were added to the wells and incubated with rocking for 1 h at rt. rSpTrf-E1 in the wells was evaluated with α -66, α -68 (2000 $\times$ dilution), and α -71 (10,000 $\times$ dilution) followed by G α Rlg-HRP (10,000 $\times$ dilution), TMB, and stop solution, and the color reaction was read at OD₄₅₀ as described above. Background readings for 0 μ g flagellin were subtracted from readings for 2 to 4 μ g flagellin.

rSpTrf incubation with flagellin or LPS in solution

Each rSpTrf protein (0.5 μ g) was mixed with LPS (5 ng; 1%), flagellin (10 ng; 2%), or without a PAMP in 40 μ L PBS and incubated for 1 h at rt. Samples were separated by 8% SDS-PAGE, transferred to PVDF filters (Immobilon-P^{seq}) for 10 min with a Trans-Blot Turbo Transfer system. Filters were blocked in blotto with rocking for 1.5 h at rt or overnight at 4 °C. Blotto was replaced and filters were incubated with α -V5-HRP antibody (3000 $\times$ dilution in fresh blotto) with rocking for 1 h at rt. Filters were washed twice in TNT and twice in TN, incubated in ECL (Super Signal West Pico PLUS) for 5 min, and evaluated for chemiluminescence in a ChemoDoc Touch Imaging System.

Structural analysis of rSpTrf proteins

Posttranslational modifications to rSpTrf proteins

Analysis of PTMs added to the rSpTrf proteins by the insect cells was carried out by the Proteomics Shared Resource Center at Georgetown University (<https://lombardi.georgetown.edu/research/sharedresources/pmsr/proteomics/>). In brief, a subset of rSpTrf proteins were digested with trypsin/Lys-C (Thermo Fisher, item A41007) and PNGaseF (New England Biolabs, item P0704S), extracted, loaded into Protifi LLC S-Trap columns (Fisher Scientific, item NC1679156), and analyzed by nanoAcquity ultra performance liquid chromatography (Waters Corp., item 12089) coupled with an Orbitrap Lumos Mass Spectrometer

(Thermo Fisher, item FETD2-10002). The mass spectrometry output was evaluated with Proteome Discoverer (Thermo Fisher) and compared to the rSpTrf protein sequences. Analyses of standard PTMs included phosphorylation of Ser, Thr, Tyr, and His, methylation of Arg and Lys, and acetylation of Asn and Lys. Deamidation of Asn to Asp by PNGaseF treatment and subsequent acetylation was used to infer sites of N-linked oligosaccharides.

Circular dichroism

CD spectra of recombinant proteins were obtained using a J-1500 CD spectrometer (Jasco Analytical Instruments, item 7000-J005A) equipped with a Peltier temperature controller. Protein samples at a concentration of 0.6 to 2.7 μM in 10 mM sodium phosphate buffer, pH 7.4, were placed in a 1-mm quartz cell. Spectra were obtained at 298 K using a 190- to 260-nm measurement range with 50 nm/min scanning speed, 1 nm bandwidth, and 8 sec response time with 1.0 nm data pitch, and were averaged over 5 scans with baseline subtraction. Baseline spectra were acquired using 10 mM sodium phosphate containing either 0 mM or 10 mM SDS in the absence of protein. CD spectra were used to calculate the molar ellipticity with units of $\text{mdeg} \times \text{cm}^2 \times \text{dmol}^{-1}$. Spectra were analyzed with Spectra Manager v2 (Jasco Analytical Instruments, item 7066-J021A).

Structure prediction with in silico tools

Secondary structure prediction based on CD spectra for each rSpTrf protein was evaluated with single-spectrum analysis for secondary structure prediction and order/disorder classification with BeStSel (<https://bestsel.elte.hu/>),^{49,50} Dichroweb (<http://dichroweb.cryst.bbk.ac.uk/html/home.shtml>),^{50–52} and K2D3 (<https://cbdm-01.zdv.uni-mainz.de/~andrade/k2d3/>).⁵³ The amino acid sequence of each rSpTrf protein was used in structural predictions by Metapredict Online (v2.4; <https://metapredict.net/>)⁵⁴ and Classification of Intrinsically Disordered Ensemble Regions (CIDER; <https://pappulab.wustl.edu/CIDER/analysis/>).^{35,55}

Predictions of secondary structure of the rSpTrf proteins were obtained from the AlphaFold 2 Protein Structure Database ($\alpha\text{F2-PDB}$, DeepMind and European Molecular Biology Laboratory–European Bioinformatics Institute; <https://alphafold.ebi.ac.uk/>).^{56–58} Structural predictions of the rSpTrf proteins by AlphaFold 3 (αF3 , DeepMind; <https://golgi.sandbox.google.com/>)⁵⁹ used the amino acid sequences of the proteins with and without the PTMs. Outputs provided confidence estimates based on a predicted local distance difference test (pLDDT)⁶⁰ and 5 predicted structure models that included the fraction disorder for each model. Structure models were viewed as ribbon structures in PyMOL (The PyMOL Molecular Graphics System, v3.0.3; Schrödinger, LLC) with parameters set for smooth loops, CMYK color space, white background, maximum quality, round and fancy helices, and flat and fancy sheets, with medium rendered shadows.

Statistical analyses

Significant differences among fraction disorder for structure models from αF3 for rSpTrf proteins with and without PTMs and for mixed pairs of proteins were determined by paired, 2-tailed *t* tests in Microsoft Excel with significance set to $P \leq 0.05$.

Results

The rSpTrf proteins from insect cells fail to bind flagellin or LPS

rSpTrf-E1-Ec binds to several PAMPs including flagellin, LPS, β -1,3-glucan,²⁶ and PA.³¹ This range of binding capabilities suggested that other SpTrf proteins may also have diverse or overlapping sets of binding targets including LPS and flagellin. To address this prediction, binding to flagellin and LPS by the rSpTrf proteins expressed in insect cells was evaluated by ELISA. Although flagellin and LPS bound to the wells of the 2HB plate (Fig. S3A–C), none of the rSpTrf proteins bound to either flagellin or LPS (Fig. 2A). Furthermore, natSpTrf proteins in cCF or isolated by IMAC also failed to bind flagellin (Fig. 2B).

Previous results have demonstrated that native Trf proteins from several sea urchin species are irreversibly multimerized upon collection (Fig. S2), which increases the molecular weight observed on western blots.^{17,18,24,28} Furthermore, when rSpTrf-E1-Ec is bound to flagellin, the complex shows increased molecular weight and both rSpTrf-E1-Ec and flagellin are present in the same band on a protein gel.²⁶ rSpTrf-E1-Ec multimerizes spontaneously and irreversibly, including when bound to PA.³¹ Consequently, to address the question of whether the rSpTrf proteins could be induced to bind PAMPs and multimerize, they were evaluated for size changes after incubating with flagellin or LPS. However, unlike rSpTrf-E1-Ec and natSpTrf proteins, the rSpTrf proteins did not show increased sizes when incubated with flagellin or LPS (Fig. 3). This suggested that the rSpTrf proteins failed to bind either flagellin or LPS, which was different from rSpTrf-E1-Ec that binds both.²⁶

The rSpTrf proteins fail to bind bacteria

Although the rSpTrf proteins did not bind flagellin or LPS, other PAMPs on bacteria may be targets. For example, rSpTrf-E1-Ec only binds Gram-negative bacteria, which may include PAMPs other than flagellin and LPS.²⁶ Furthermore, natSpTrf proteins bind Gram-negative bacteria *V. diazotrophicus*, Gram-positive bacteria *B. cereus*, and the yeast *S. cerevisiae*, in addition to a range of foreign cells³² (see the supplementary file in Lun et al.²⁶). When the natSpTrf proteins are secreted from phagocytes in short-term cultures, they bind to *V. diazotrophicus* and augment phagocytosis.²⁹ Consequently, the rSpTrf proteins from insect cells were evaluated for binding to bacteria by In-cell ELISA. *Vibrio diazotrophicus* and *B. cereus* were fixed to wells, as verified by an anti-flagellin antibody (Fig. S3D), and incubated with the rSpTrf proteins. Results showed that none of the proteins bound to either species of bacteria (Fig. 2C), which was similar to binding by the irrelevant control protein, frog rCSF-1-V5, that was produced in the same insect cell line.^{46,47} This was in disagreement with natSpTrf proteins that bind intact bacteria.^{26,32} The rSpTrf proteins are modified with N-linked oligosaccharides,³⁷ consequently to determine whether they interfered with binding to PAMPs, rSpTrf-E1 was treated with PNGaseF and evaluated for binding flagellin. Comparisons between deglycosylated and glycosylated proteins showed that the removal of N-linked oligosaccharides did not improve binding to flagellin (Fig. 2D). These results suggested that the rSpTrf

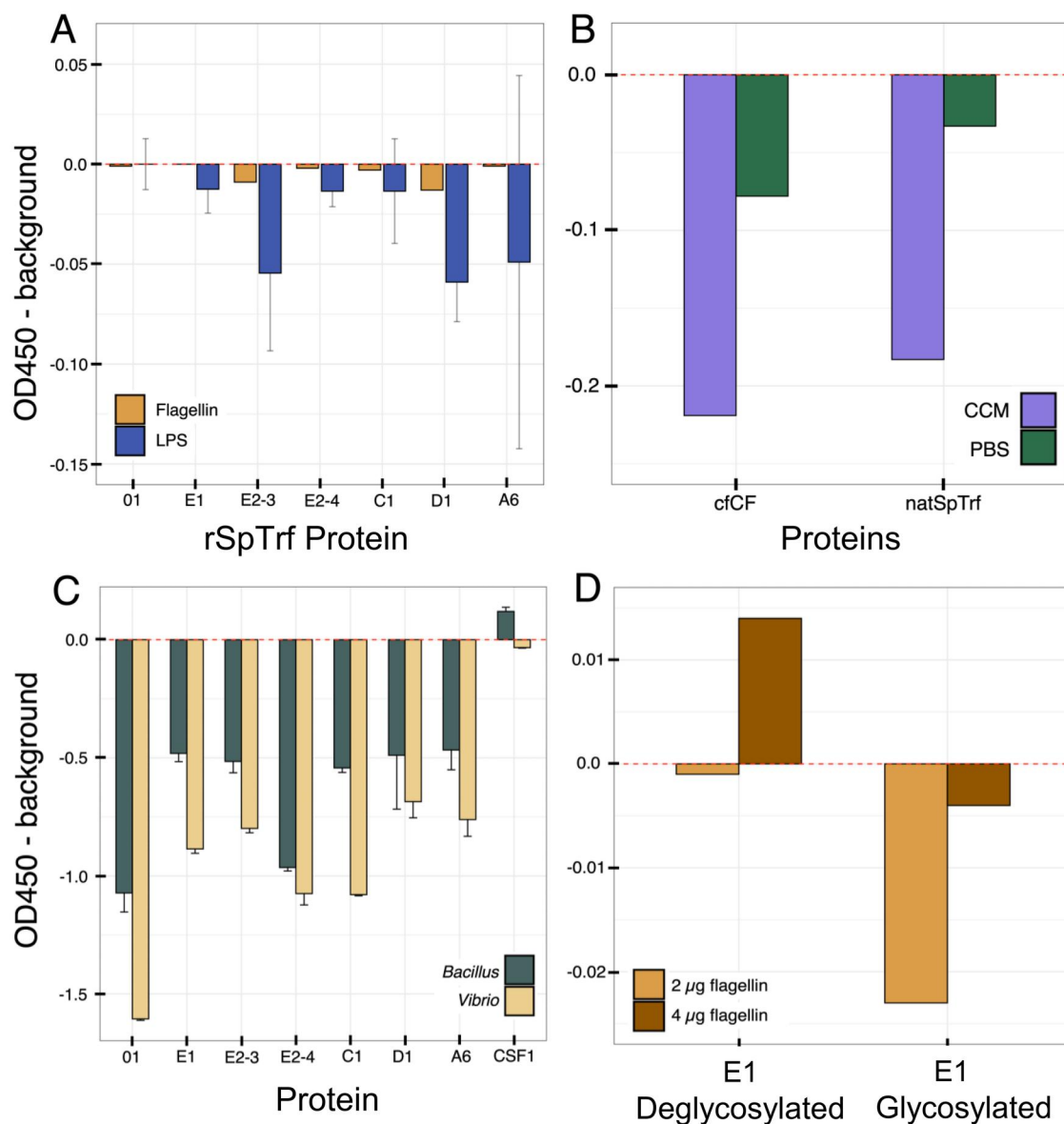


Figure 2. rSpTrf and natSpTrf proteins fail to bind PAMPs or bacteria. (A) The rSpTrf proteins fail to bind flagellin or LPS. Flagellin and LPS were bound to wells of 2HB plates (Fig. S3A-C) and rSpTrf proteins were evaluated for binding the PAMPs by ELISA. Background binding was determined for wells without PAMPs, which was subtracted from wells with PAMPs. An additional control omitted rSpTrf proteins from wells to evaluate binding by the α -V5-HRP antibodies with results similar to rSpTrf protein binding to wells without PAMPs. (B) Cell-free coelomic fluid (cfCF) with natSpTrf proteins or IMAC-isolated natSpTrf proteins fail to bind flagellin. natSpTrf protein binding to flagellin in wells was evaluated by ELISA with 3 rabbit-anti-natSpTrf antibodies (α -66, α -68, α -71) followed by G α Rlg-HRP. Binding was tested in both PBS and coelomocyte culture medium (CCM).⁴⁸ Background binding by the proteins was determined from wells without flagellin and results are shown after background subtraction. (C) rSpTrf proteins fail to bind bacteria. *Vibrio diazotrophicus* or *Bacillus cereus* were bound to wells (Fig. S3D) and used to evaluate rSpTrf protein binding by In-cell ELISA with the α -V5-HRP antibody. A recombinant frog cytokine, rCSF-1-V5,⁴⁶ served as the negative control. Background binding by the proteins was determined from wells without bacteria and results are shown after background subtraction. (D) Deglycosylated rSpTrf-E1 does not enable binding to flagellin. Deglycosylated and glycosylated rSpTrf-E1 (0.55 μ M) were evaluated by ELISA for binding to flagellin in wells (Fig. S3A) using 3 anti-natSpTrf antibodies followed by G α Rlg-HRP. Results are shown after subtraction of the background controls that omitted flagellin from the wells. The y axes in (A) and (C) apply to (B) and (D).

proteins did not bind to structures on Gram-positive and Gram-negative bacteria and that this failure may not involve the N-linked oligosaccharides. This was in agreement with the failure of the IMAC-isolated natSpTrf proteins to bind flagellin (Fig. 2B) and was in contrast to previous findings for rSpTrf-E1-Ec that is not glycosylated and binds flagellin and Gram-negative *V. diazotrophicus*.^{26,29}

The anti-natSpTrf antibodies bind variably to the rSpTrf proteins

The rSpTrf proteins are modified by the insect cells with the addition of N-linked oligosaccharides.³⁷ However, it was not known whether N-linked oligosaccharides may have altered the protein structure and thereby inhibited binding to PAMPs or

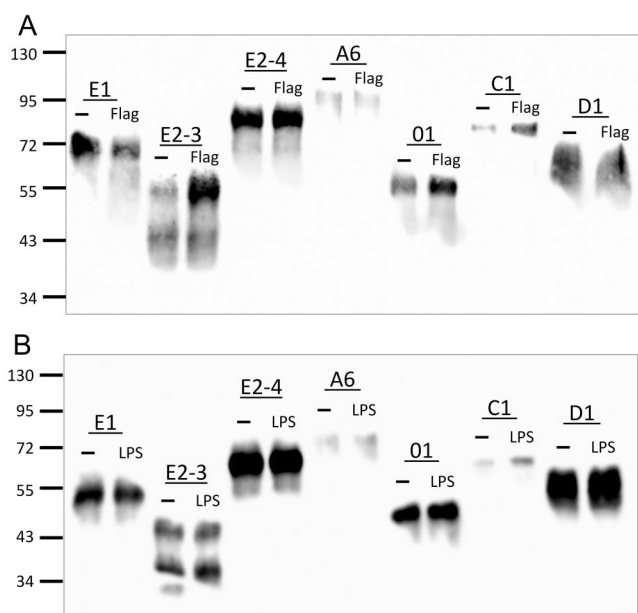


Figure 3. rSpTrf proteins mixed with PAMPs do not show a size shift. Each rSpTrf protein was incubated (A) with or without flagellin (flag) or (B) with or without LPS. Evaluation was carried out by western blot based on detection with α -V5-HRP. Protein standard sizes (kDa) are shown to the left.

bacteria. Given the different binding outcomes for the rSpTrf proteins compared to rSpTrf-E1-Ec, 3 rabbit antibodies, α -66, α -68, and α -71, were tested for binding to the rSpTrf proteins to infer possible changes to the protein structures. The binding sites for these antibodies are in conserved regions of SpTrf protein sequences located near the N-terminus, within the multimerization region, and within the C-terminal region (Fig. S1).^{24,28} Results from ELISA showed that the 3 antibodies bound differently among the 6 rSpTrf proteins with particularly poor binding by α -66 and α -68 (Table 1, Fig. S4). In comparison, the 3 antibodies bound equally well to the natSpTrf proteins. These results supported the notion that N-linked oligosaccharides were at least near the binding sites for α -66 and α -68 and may have interfered with antibody binding. To test this, a subset of proteins, rSpTrf-E1, -E2-3, and -E2-4, were deglycosylated and evaluated for binding by the anti-natSpTrf antibodies. Results showed that antibody binding by α -66 and α -68 was not improved for deglycosylated proteins, which was not different from glycosylated and untreated proteins (Table 2, Fig. S5). N-linked oligosaccharides that were added to the rSpTrf proteins by the insect cells were not the basis for the reduced binding by α -66 and α -68 in the N-terminal and multimerization regions. Furthermore, it is noteworthy that the natSpTrf proteins are glycosylated by phagocytes,³⁷ but these PTMs did not impact binding by the anti-natSpTrf antibodies. Overall, these results suggested that the rSpTrf proteins may be different in their overall structure compared to the natSpTrf proteins and that this difference may restrict access of the antibodies to their binding sites.

The rSpTrf proteins are disordered

rSpTrf-E1-Ec is not posttranslationally modified by *E. coli* and is an intrinsically disordered protein (IDP) of 99% disordered structure. It transforms to approximately 80% α -helical when incubated with

Table 1. Anti-natSpTrf antibodies bind variably to different rSpTrf proteins^a

rSpTrf protein	Anti-natSpTrf antibodies ^b [binding site ^c]		
	α -66 [N-terminal region]	α -68 [multimerization region]	α -71 [C-terminal region]
-01	— ^d	—	+
-E1	++	++	++++
-E2-3	++	++	++++
-E2-4	—	—	+++
-C1	—	—	++
-D1	++++	++++	+++
-A6	—	—	—
natSpTrf	++	+	+

^a Antibody binding to 2.0 nM rSpTrf proteins and 50 ng natSpTrf proteins were evaluated by ELISA. See Fig. S4.

^b For details, see Brockton et al.²⁸ and Fig. S1.

^c The binding site regions are shown in Fig. 4.

^d —, no binding; +, low binding; ++, medium binding; +++, high binding; +++++, very high binding.

Table 2. Glycosylation on the rSpTrf proteins does not interfere with anti-natSpTrf antibody binding^a

rSpTrf protein	Treatment	Anti-natSpTrf antibodies ^b		
		α -66	α -68	α -71
-E1	Deglycosylated	+ ^c	+	++
	Glycosylated	+	+	++
	Untreated	+	++	++
-E2-3	Deglycosylated	+	+	+++
	Glycosylated	++	+	+++
	Untreated	++	++	+++
-E2-4	Deglycosylated	—	—	+
	Glycosylated	—	—	++
	Untreated	—	—	+

^a Antibody binding was determined by ELISA. See Fig. S5.

^b For details, see Brockton et al.²⁸ and Fig. S1.

^c —, no binding; +, low binding; ++, medium binding; +++, high binding.

LPS, 2,2,2-trifluoroethanol, the anionic detergent SDS, or the phospholipid PA.^{27,31} Because the structures of the rSpTrf proteins isolated from insect cells were unknown and may have precluded binding to PAMPs and bacteria, they were evaluated by CD. Scans showed that each of the rSpTrf proteins was disordered (Fig. 4A) with identical CD spectra reported for rSpTrf-E1-Ec.²⁷ Additional analyses by Metapredict Online (<https://metapredict.net/>) for the SpTrf proteins identified in the α F2-PDB (<https://alphafold.ebi.ac.uk/>) predicted disorder throughout the SpTrf proteins except for the α -helical leader (Fig. S7), which was not included in the expression clones for either insect cells or *E. coli*.^{26,37} To verify the percent disorder of each protein, the CD spectral data were evaluated by BeStSel (<https://bestsel.elte.hu/index.php>),⁴⁹ which showed that each protein had a similar predicted structure of approximately 0% helix, 15% turns, and approximately 50% “other,” suggesting approximately 65% disorder (Fig. 4B). Although the predictions for

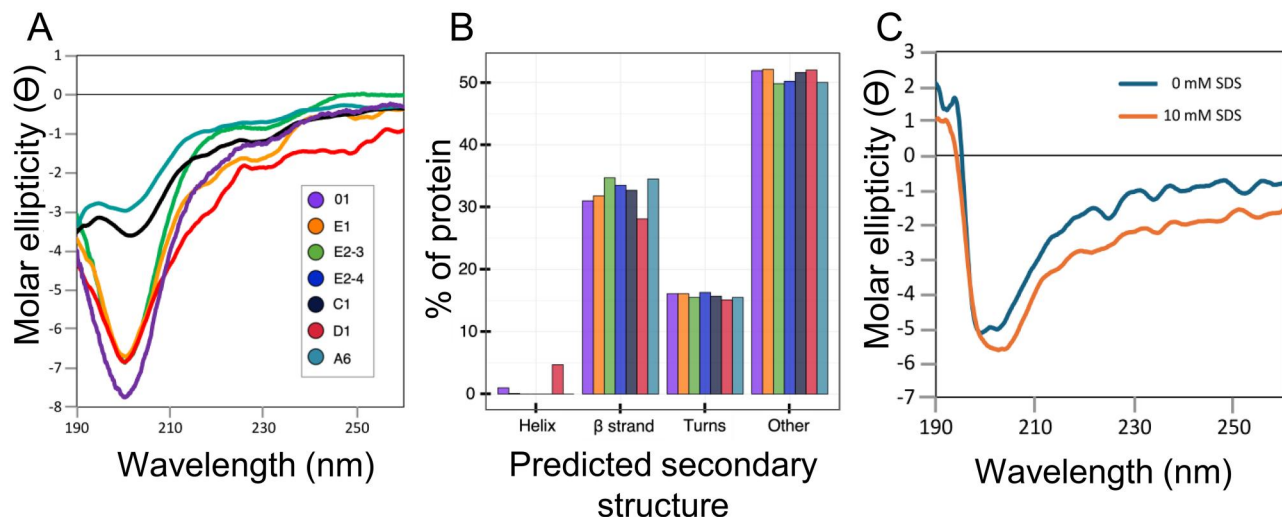


Figure 4. The rSpTrf proteins are disordered with predicted β -strands and do not transform with the addition of SDS. (A) The CD spectra suggests that all of the rSpTrf proteins are disordered in agreement with rSpTrf-E1-Ec.²⁷ Variations in the scans relative to the y axis are due to differences in protein concentration. (B) The CD spectral data from the results in (A) were evaluated with BeStSel (<https://bestsel.elte.hu/>) that predicts secondary structure of 28% to 34.5% β -strands and essentially no helices for each rSpTrf protein. The legend in (A) applies to (B). (C) rSpTrf-E1 was evaluated by CD for transformation to α -helical secondary structure in 10 mM SDS, as reported for rSpTrf-E1-Ec.²⁷ The protein spectral results with and without SDS are not different, indicating that rSpTrf-E1 does not undergo a transformation from disorder to α -helical. The scans for the samples of rSpTrf-E1 in (A) and (C) appear different because they were isolated at different times in separate batches. Molar ellipticity (Θ) is based on sample concentration (gm/L), light pathlength in the instrument in cm, and protein molecular weight or (deg cm²/dmol) $\times 10^3$.

random coil or disorder encompassed 50% to 65% of the protein sequences, this was not in agreement with the CD results for 99% disordered structure of rSpTrf-E1-Ec.²⁷ Unexpectedly, results also showed 30% to 35% β -strands with predictions of approximately 48% to 58% β -strands by DichroWeb (<http://dichroweb.cryst.bbk.ac.uk/html/home.shtml>)⁵⁰ and K2D3 (<https://cbdm-01.zdv.uni-mainz.de/~andrade/k2d3/>)⁵³ (Fig. S6). These predictions of β -strands correlated with a CD analysis of a Gly-rich region recombinant fragment of rSpTrf-E1-Ec that transforms to β -strands when LPS and other reagents are added to the solution.²⁷ rSpTrf-E1-Ec undergoes a transformation in the presence of several types of binding targets including SDS.^{27,31} Because rSpTrf-E1 from insect cells has the identical sequence as rSpTrf-E1-Ec and appeared disordered based on results from CD, α F2, and α F3, rSpTrf-E1 was evaluated by CD for transformation to α -helical when mixed with SDS. However, the structure before and after the addition of SDS was not different, indicating that the protein did not change conformation from disordered to any type of secondary structure (Fig. 4C). The failure of rSpTrf-E1 to undergo a structural transformation when mixed with SDS was unlike rSpTrf-E1-Ec, suggesting a fundamental difference between the recombinant proteins produced in insect cells compared to the protein produced in *E. coli*.

The rSpTrf proteins show variable posttranslational modifications

The inability of the rSpTrf proteins to bind PAMPs and bacteria and undergo conformational changes, and the failure of rSpTrf-E1 to undergo transformation with SDS, did not align with the results for rSpTrf-E1-Ec.^{26,27,31} This suggested that PTMs other than N-linked oligosaccharides may have been added by the

insect cells that induced structural changes and altered the range of rSpTrf protein functions. The locations of putative PTMs were predicted by in silico analysis (Fig. S1); however, the actual PTMs that were added by insect cells and their locations were unknown. Therefore, a subset of rSpTrf proteins were evaluated by proteomics to identify the types and locations of PTMs. Proteins chosen for analysis included (i) rSpTrf-E1 (from insect cells), (ii) rSpTrf-E2-3 (monomer) compared to rSpTrf-E2-4 (dimer), and (iii) rSpTrf-A6 compared to rSpTrf-D1 (based on differences in antibody binding) (Table 1, Fig. S4; also see Fig. 2C in Crow et al.³⁷). In general, the proteomic results showed that the insect cells modified the rSpTrf proteins with several different types of PTMs located in a variety of positions, and that there was no PTM in a shared position on all proteins (Fig. 5, Table 3, Fig. S1). The number of PTMs on the rSpTrf proteins ranged from 4 to 13. Results from all proteins combined showed that there were 16 Asn positions to which N-linked oligosaccharides were added, whereas in silico tools predicted that 22 Asn had N-linked oligosaccharides, of which 11 were correct, and 5 that were not predicted (Fig. S1). Notably, there was diversity in the numbers and locations of PTMs in the multimerization region with methylated arginine (Me-R) added to some proteins but not to others (Fig. 5). The PTMs in the His-rich region also showed diversity in which some proteins had several PTMs, whereas others had none.

PTMs are added to the rSpTrf-E1 protein

rSpTrf-E1 had the most PTMs among the proteins that were evaluated with 7 in the Gly-rich region, 6 in the His-rich region, and none in the C-terminal region including the V5 tag binding site (Fig. 5, Fig. S1). In the Gly-rich region, an N-linked oligosaccharide was located within the α -66 binding site, and an Me-R was located

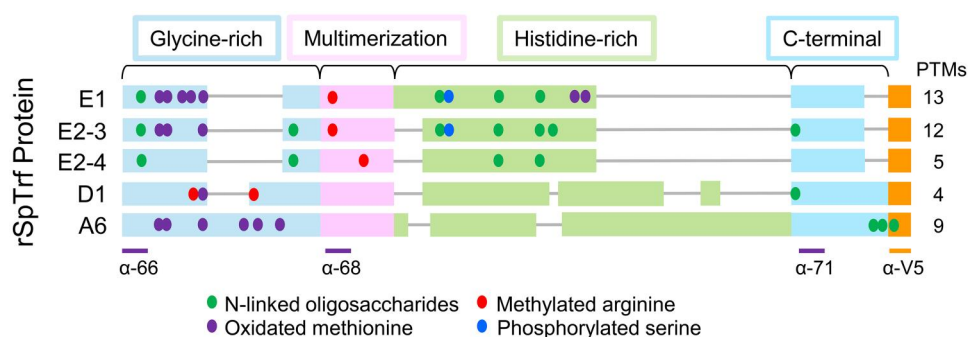


Figure 5. The rSpTrf proteins have several PTMs in inconsistent locations. PTMs added to the rSpTrf proteins by the insect cells are indicated based on proteomic analyses. The proteins have variations in PTM type and location. The regions of a standard SpTrf protein are indicated at the top in addition to the V5 and His tags at the C-terminus (orange). The total number of PTMs on each protein is indicated to the right. The binding sites for the 3 rabbit anti-natSpTrf antibodies (α -66, α -68, α -71) and the α -V5 antibody are indicated at the bottom of the alignment. This alignment is different from that shown in Fig. 1 because it was generated de novo with the 5 protein sequences. See Fig. S1 for the details of protein sequences and PTM locations.

within the α -68 binding site and appeared to correlate with differences in α -66 and α -68 antibody binding (Table 1). In general, rSpTrf-E1 did not show secondary structural transformation when mixed with SDS compared to rSpTrf-E1-Ec without PTMs that transformed from disordered to α -helical.

Different PTMs are added to the rSpTrf-E2 proteins

The 2 rSpTrf-E2 proteins have identical sequences; however, based on size, rSpTrf-E2-3 is a monomer and rSpTrf-E2-4 is an irreversible dimer.³⁷ rSpTrf-E2-3 had 12 PTMs whereas rSpTrf-E2-4 had 5 (Fig. 5, Table 3). The 2 proteins shared locations for 4 of 7 N-linked oligosaccharides with 2 additional sites in rSpTrf-E2-3 in the His-rich region and one in the C-terminal region. Both proteins had N-linked oligosaccharides positioned within the α -66 binding site, whereas rSpTrf-E2-3 also had a PTM adjacent to the α -71 binding site (Fig. S1). Other differences included 3 oxidized methionines (Ox-M) in the Gly-rich region that were absent from rSpTrf-E2-4, the locations of Me-R in the multimerization region, and 5 PTMs in the His-rich region of rSpTrf-E2-3 but only 2 in rSpTrf-E2-4. The locations of N-linked oligosaccharides in these 2 proteins correlated with reduced binding by the α -66 and α -68 antibodies (Table 1, Figs. S1 and S4A, B).

PTMs on rSpTrf-D1 are different from those on rSpTrf-A6

rSpTrf-D1 and -A6 were compared because of differences in antibody binding (Table 1). rSpTrf-D1 had only a total of 4 PTMs, the fewest compared to the other rSpTrf proteins (Fig. 5). In comparison, rSpTrf-A6 had 9 PTMs, of which 6 Ox-M were located in the Gly-rich region. There were no PTMs positioned within the binding sites for α -66 or α -68 for either protein. This positioning correlated with good binding by these antibodies, which differed from rSpTrf-E1 and both rSpTrf-E2 proteins (Table 1, Figs. S1 and S4A, B). Unlike the other proteins, rSpTrf-A6 had one N-linked oligosaccharide in the α -V5 binding site (Fig. 5, Table 3) that correlated with poor visualization on western blots (Fig. 3). Alternatively, the numbers and positions of PTMs on rSpTrf-D1 correlated with strong visualization with α -V5 on western blots (Fig. 3; also see Crow et al.^{30,37}) and strong detection by the 3 anti-natSpTrf antibodies by ELISA (Table 1, Fig. S4). Overall, the insect cells added several PTMs to the rSpTrf proteins with

differences in the types, numbers, and locations among the proteins, including variations between the 2 rSpTrf-E2 proteins of identical sequence. These additions correlated with level of the detection by the antibodies and the failure of the proteins to undergo structural transformation.

Predictions for the rSpTrf proteins include disorder and some secondary structure

rSpTrf-E1-Ec is disordered based on predictions from in silico tools and results from CD.^{26,27} The disordered structure of the SpTrf proteins from insect cells has also been predicted by α F2 (accessed April 2023) (Fig. S7). To predict any structural differences for the rSpTrf proteins with PTMs, they were reevaluated with α F3 (accessed May–August and December 2024) that allowed sequence analysis with some PTMs (Ox-M was not an option in α F3). Although the AlphaFold algorithm tends to be more accurate for predicting structures of ordered proteins,⁵⁶ the pLDDT from α F3 indicated low to very low confidence scores for the output structure predictions for each rSpTrf protein. This result can be interpreted as a reliable prediction that these proteins are IDPs.⁶¹ Low confidence scores also infer structural ensembles of IDPs, which are heterogeneous sets of conformations for individual proteins⁶² that undergo a continuum of conformational dynamics or “foldons” (reviewed in Uversky,³⁴ Gupta and Uversky,⁶³ and Holehouse and Kragelund⁶⁴). Consequently, the computational predictions of IDP conformation by α F3 that is based on the primary sequence cannot be interpreted as a static protein structure.⁶¹ Furthermore, evaluation of the rSpTrf protein sequences with the online tool CIDER (v1.7; <http://pappulab.wustl.edu/CIDER/analysis/>) categorized the proteins as Janus proteins with unpredictable structures of both disordered or folded depending on the chemical context of the milieu.^{35,55,65} Nevertheless, given these caveats for IDPs, a few results were noteworthy for structure predictions of the rSpTrf proteins by α F3. Decreased disorder, or increased secondary structure, was predicted consistently for the rSpTrf proteins when PTMs were included in the analysis (Fig. 6, Fig. S8). Similarly, a significant decrease in disorder was predicted when the estimates of fraction disorder⁶⁶ associated with the 5 structure models from α F3 were used to compare the rSpTrf proteins with and without PTMs

Table 3. Posttranslational modifications are present in each of the rSpTrf proteins^a

Regions, sites, and numbers of PTMs within the rSpTrf proteins ^b									
rSpTrf protein	Gly-rich region ^b	α-66 site ^a	Multimer region ^b	α-68 site ^a	His-rich region ^b	C-terminal region ^b	α-71 site ^a	V5 tag	Total PTMs
E1	5 Ox-M	1 N-linked oligosaccharide ^c	1 Ox-M	1 Me-R	3 N-linked oligosaccharide 1 PO ₄ -S 2 Ox-M	— ^d	—	—	13
E2-3	3 Ox-M 1 N-linked oligosaccharide	1 N-linked oligosaccharide		1 Me-R	3 N-linked oligosaccharide 1 PO ₄ -S	1 N-linked oligosaccharide ^e	—	—	11
E2-4	1 N-linked oligosaccharide	1 N-linked oligosaccharide	1 Me-R ^f		2 N-linked oligosaccharide	—	—	—	5
D1	2 Me-R 1 Ox-M	—	—	—	—	1 N-linked oligosaccharide ^e	—	—	4
A6	6 Ox-M	—	—	—	—	2 N-linked oligosaccharide	—	1 N-linked oligosaccharide	9

^a These results are also shown in alignments that indicate the locations of the PTMs in rSpTrf proteins (Fig. 5, Fig. S1). The alignments also show the locations and sequences of the antibody binding sites.

^b See Fig. 1 for the general structure of the SpTrf proteins.

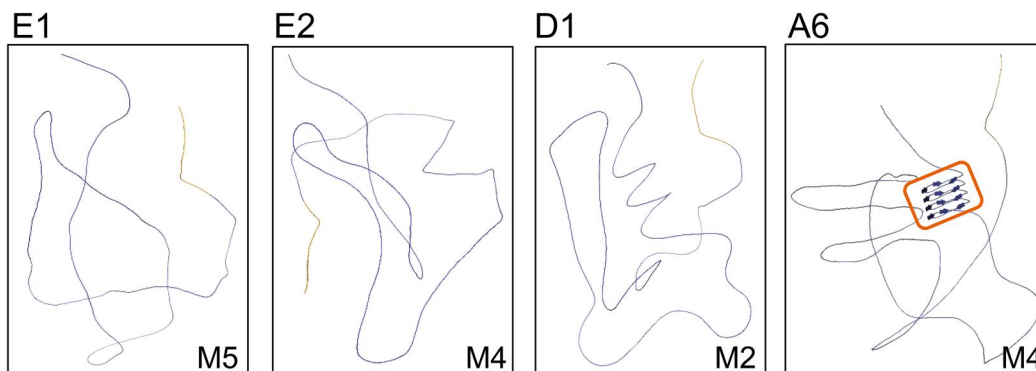
^c N-linked oligosaccharides are inferred from deamidated and acetylated Asn after treatment with PNGaseF.

^d —, no PTMs are identified in this site or region of the protein.

^e This N-linked oligosaccharide is located immediately to the N-terminal side of the α-71 binding site.

^f This Me-R is located 3 amino acids to the C-terminal side of the α-68 binding site.

A Structural predictions of rSpTrf proteins



B Structural predictions of proteins with PTMs

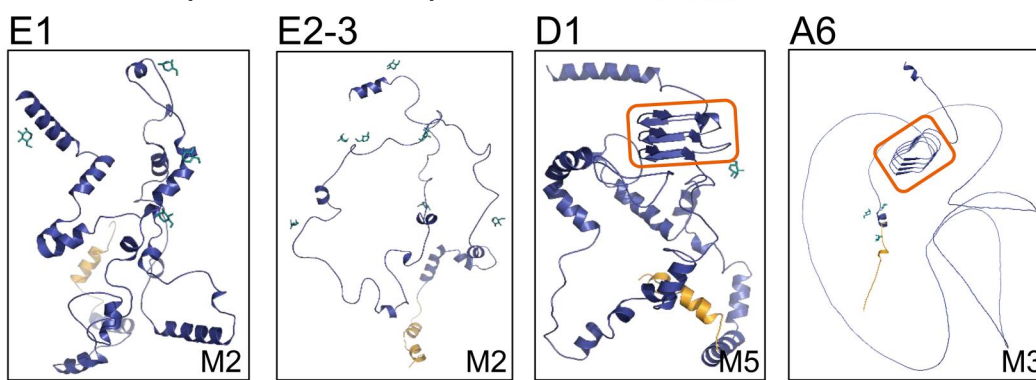


Figure 6. Predicted rSpTrf protein structures indicate disorder with increased secondary structure with PTMs. A selected set of rSpTrf proteins illustrate the structure predictions from α F3. (A) Structure predictions based on the protein sequences without PTMs. (B) Structure predictions based on the protein sequences including PTMs. See Fig. S8 for the structure predictions for all proteins. The N-terminus is generally shown toward the top left of each panel, the V5 and 6His tags (yellow) are located at the C-terminus, and the locations of the N-linked oligosaccharides are indicated (cyan). Predicted structures suggest a β solenoid (orange boxes) in the Gly-rich region (see Fig. 1). α F3 provides 5 models of predicted structures for each protein and the models for illustration are listed at the bottom of each panel. Models were selected for optimal representation in 2 dimensions based on visualization in PyMOL (v3.0.3). The model numbers (M1 to M5) correspond to the model numbers for fraction disorder of the proteins in Fig. 7A and B.

(Fig. 7A, B, Fig. S8). These predictions suggested increased secondary structure of the rSpTrf proteins, which were consistent with increased stability.³⁷ These predictions were different from rSpTrf-E1-Ec that is disordered, unmodified, very unstable, and prone to spontaneous multimerization.^{26,31} Although structural predictions of IDPs by α F3 cannot be construed as stable conformational changes in the proteins, the overall results for the rSpTrf proteins suggested some type of increased secondary structure with the addition of PTMs.

Pairs of different rSpTrf proteins predict decreased structure disorder

A wide variety of similar natSpTrf proteins are secreted into the sea urchin CF by the phagocytes, and these proteins have been predicted to function together to opsonize foreign particles and/or to multimerize irreversibly^{26,30,31,37} (Fig. S2). Consequently, α F3 was used to predict whether pairs of different SpTrf proteins might interact and alter their structures. All combinations of protein pairs resulted in similar general predictions of reduced fraction disorder or increased secondary structure compared to proteins evaluated alone (Fig. 7C–F, Fig. 8, Fig. S9). The addition of

PTMs further decreased the predicted fraction disorder and correlated with significantly increased secondary structure predictions. The interactions between different pairs of proteins correlated with multimerization observed for both natSpTrf proteins and rSpTrf-E1-Ec,^{26,28,31} in addition to the stability of the rSpTrf proteins from insect cells as described here and previously.^{30,37} Although the specific structures predicted by α F3 for pairs of rSpTrf proteins cannot be taken as definitive nor can their structures be assumed to be static, the general prediction of increased secondary structure for pairs of proteins with PTMs was consistent with the characteristics of both the native and recombinant SpTrf proteins.

Discussion

rSpTrf-E1-Ec and rSpTrf proteins have opposite functions for binding foreign targets versus binding phagocytes

The functions of rSpTrf-E1-Ec (reviewed in Smith and Lun¹¹) are not consistent with the functions of the rSpTrf proteins

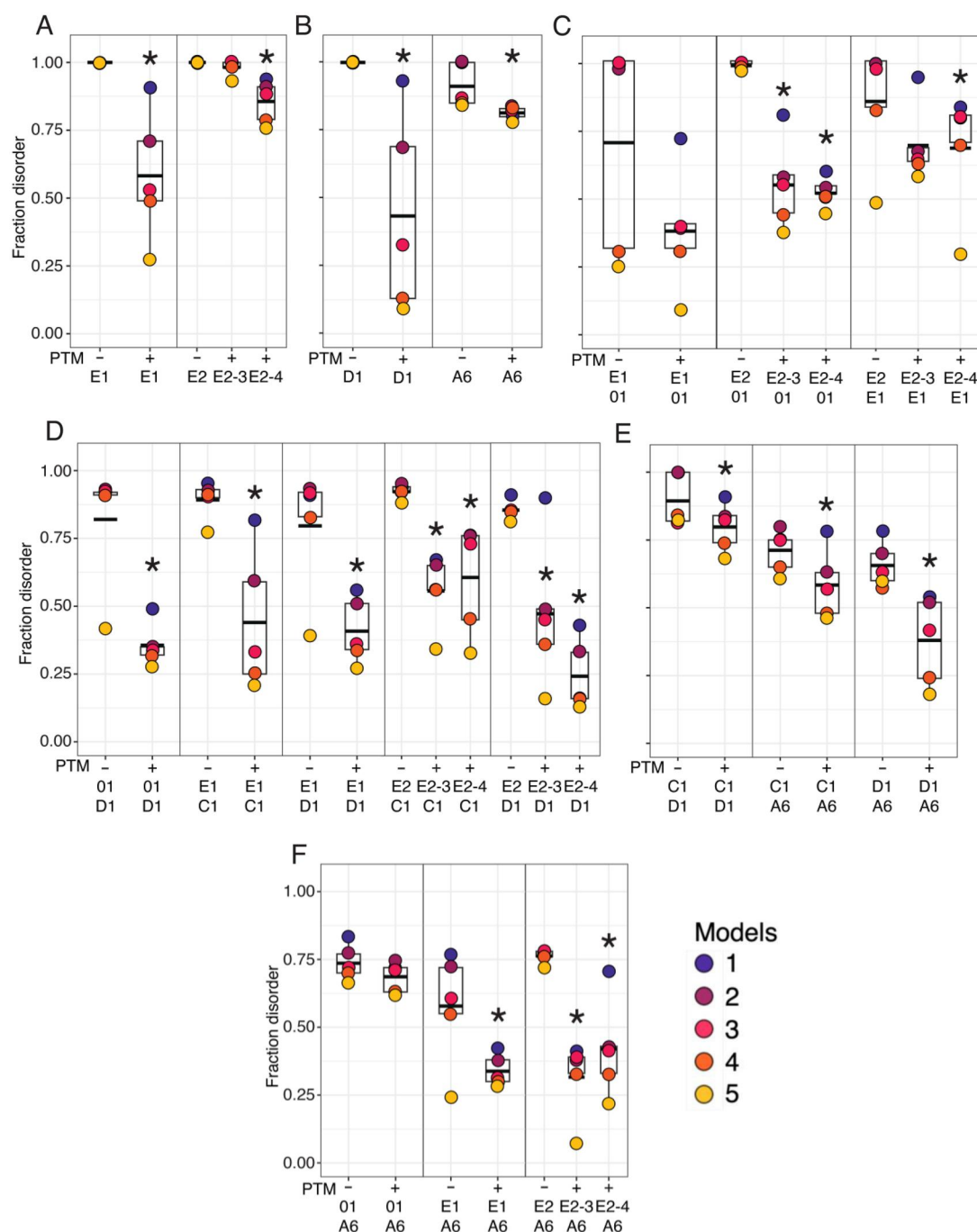


Figure 7. Fraction disorder based on structure predictions is reduced for pairs of different rSpTrf proteins with PTMs. Estimates of fraction disorder for all models of predicted structures provided by $\alpha F3$ (see legend) is used for comparisons of individual proteins and pairs of proteins with and without PTMs. (A) The rSpTrf-E type proteins. (B) rSpTrf-D1 and -A6 proteins. The fraction disorder for rSpTrf-01 and -C1 without PTMs is 100% and are not shown because they were not evaluated for PTMs. (C-F) Pairs of different rSpTrf proteins show significantly decreased fraction disorder when evaluated with PTMs. Predictions of fraction disorder of individual or pairs of different proteins is significantly decreased when evaluated with PTMs (paired t test, $*p \leq 0.05$). The legend for models applies to all panels and correlates with model numbers in Fig. 6 and Figs. S8 and S9.

produced by insect cells reported here and previously (Table 4).^{30,37} rSpTrf-E1-Ec is not modified with PTMs, is unstable, readily multimerizes, and loses binding function as a dimer or multimer.^{26,31} Monomeric rSpTrf-E1-Ec binds multiple targets including PAMPs, Gram-negative bacteria, yeast, the phospholipid PA and other lipids with similarly exposed phosphate

groups, and other similar structures.^{26,31} Although rSpTrf-E1-Ec binds to *V. diazotrophicus*, it does not augment phagocytosis by sea urchin phagocytes.²⁹ These functions suggest that its conformation as an IDP enables binding to a range of foreign targets, but once bound its putative structural transformation and/or its lack of PTMs and associated secondary structure do not

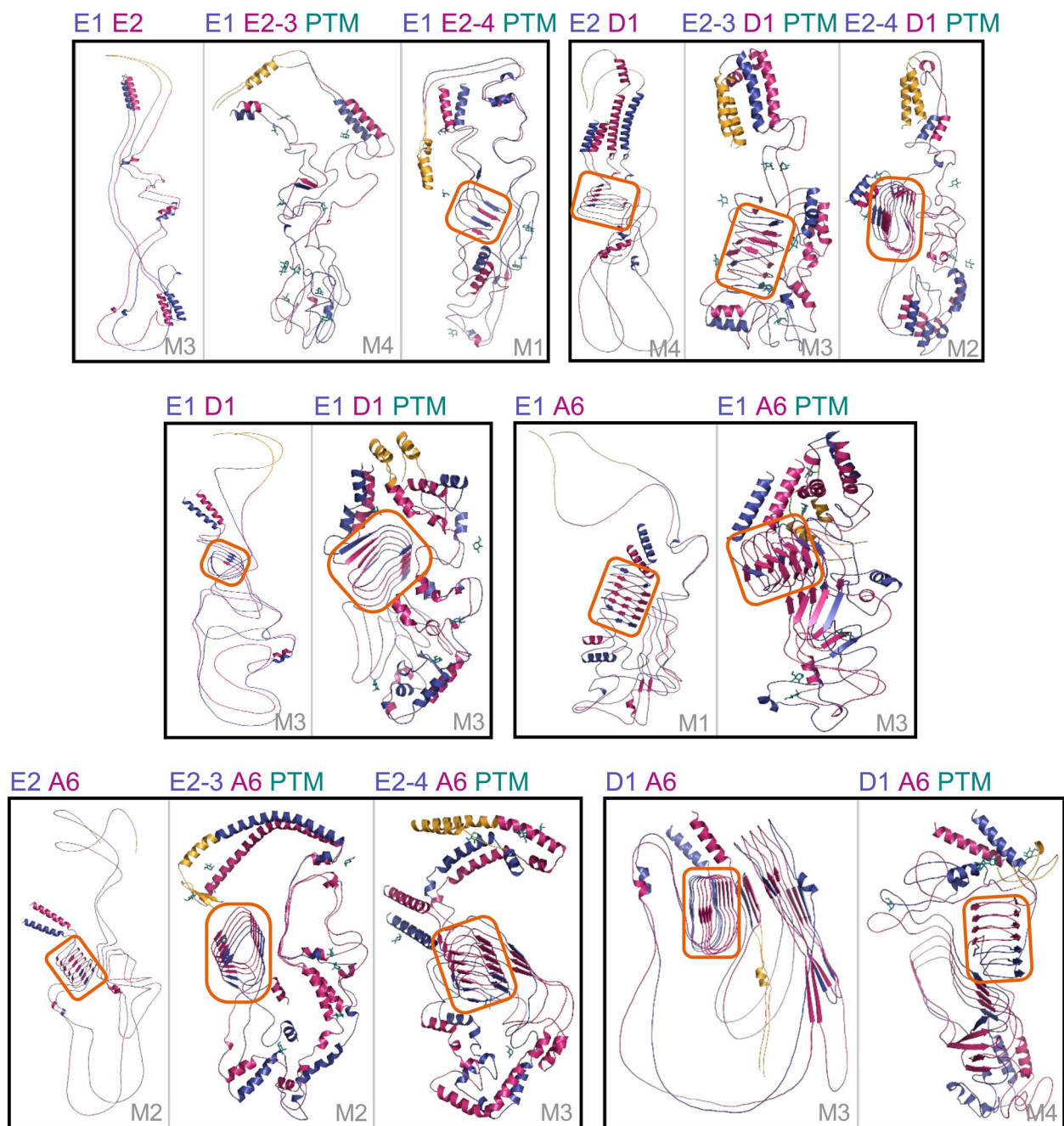


Figure 8. Pairs of different proteins with PTMs result in predictions of increased secondary structure. A selected set of protein pairs with and without PTMs illustrate structure predictions by α F3. Each panel is labeled at the top left with the shorter protein in blue and the longer protein in dark pink. The N-termini of the proteins are generally oriented toward the top left of each panel. Also shown are the V5 and 6His tags (yellow) at the C-terminus, the N-linked oligosaccharides (cyan), and the β solenoid (orange boxes) in the Gly-rich region of the proteins. The model number of the predicted structures from α F3 for each protein pair is indicated in the lower right corner of each panel and corresponds to model numbers in Fig. 7C–F. See Fig. S9 for structure predictions for all pairs of proteins.

enable recognition by phagocytes for target removal. In contrast, insect cells add PTMs to the rSpTrf proteins, which are stable, do not multimerize, do not bind PAMPs or bacteria, but bind phagocytes and augment phagocytosis by small and polygonal phagocytes when cross-linked to inert beads.^{30,37} These findings indicate significant differences in protein structure and function based on the presence, location, type of PTMs, and the overall putative secondary structure of the proteins.

Western blots show that the natSpTrf proteins in cfCF or isolation by IMAC are larger than sizes deduced from the cDNA sequences, which is consistent with irreversible multimerization. The natSpTrf proteins bind to a range of bacterial species,^{26,32} they are secreted from phagocytes, and they opsonize *V. diazotrophicus* to augment phagocytosis.²⁹ However, the natSpTrf proteins fail to bind flagellin, which may be the outcome of multimerization that results in deactivation and occurs during

Table 4. rSpTrf proteins from *Escherichia coli* and insect cells have opposite functions.

Attribute	rSpTrf-E1-Ec ^a	rSpTrf ^b
Expression	<i>E. coli</i>	Insect cells
Disorder	99%	48%–69%
α -Helical	1%	0–4%
β -Strand	0%	32%–39%
Stability	—	+
PTMs	—	+ ^c
Target binding	+	—
Transformation	+	—
Multimerization	+	—
Phagocyte binding	—	+
Opsonization	—	+
Phagocytosis	—	+
Gene regulation	?	+

^a These results are reported in Lun et al.^{26,27,31} and Chou et al.²⁹

^b These results are reported in Crow et al.^{30,37} and this paper.

^c PTMs added by insect cells are likely different from those added by phagocytes.

collection and isolation. It is noteworthy that this is similar to the failed binding activities of the rSpTrf proteins from insect cells and may be related to overall protein structure. Based on these findings, we propose that the range of functions identified for recombinant proteins isolated from *E. coli* and from insect cells are representative of the full range of functions carried out by the native proteins in vivo. This includes binding to and opsonizing foreign targets,²⁶ structural transformation (this paper),^{27,31} and binding to phagocytes that augments phagocytosis.^{29,30,37}

SpTrf PTMs impact binding foreign targets versus binding phagocytes

There are differences in the cell-binding capacities of the several rSpTrf proteins from insect cells,³⁰ which may be based on their sequence diversity, structural differences, their conformational transformation, and/or the presence and locations of PTMs. It is likely that the PTMs added by insect cells are distinct compared to those added by sea urchin phagocytes with regard to both types and locations, which is consistent with variations in binding by the 3 anti-natSpTrf antibodies to the recombinant versus native proteins. Differences in type and location of PTMs may have direct impacts on possible secondary structures and overall shape of the proteins. This may include their ability to transform from a disordered to more ordered structure that may promote, alter, restrict, or block interactions with foreign targets. Alternatively, structural variations among the native proteins may enable recognition of a wide range of foreign targets followed by transformation and binding to phagocytes. The types and location of PTMs added by insect cells may be sufficient for recognition by and binding to phagocytes. A key aspect of the PTMs and protein-binding capacities is illustrated by rSpTrf-E1-Ec and rSpTrf-E1 from insect cells that share identical sequence. rSpTrf-E1-Ec without PTMs binds non-self and undergoes a structural transformation, but is not recognized by

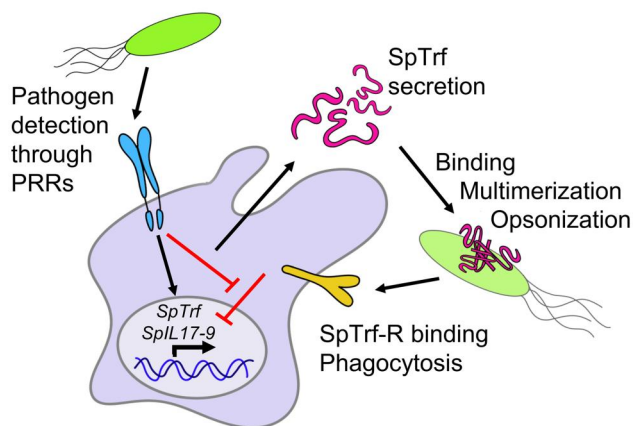
phagocytes.^{26,27,31} In contrast, rSpTrf-E1 with PTMs fails to bind non-self, does not undergo structural transformation, and may have reduced disorder, but binds to a putative SpTrf receptor (SpTrf-R) on phagocytes.^{30,37} Overall, these differences among the recombinant proteins suggest that both structural conformation plus the type and location of PTMs may impact the functions of both the native and recombinant SpTrf proteins. Future rigorous analysis of the PTMs on natSpTrf proteins may define structure–function relationships for these dynamically disordered proteins.

Conclusion

The native and recombinant SpTrf proteins, their functions in vitro, and their predicted dynamic structural changes provide a hypothetical basis for how the natSpTrf proteins may function in the sea urchin immune system in vivo (Fig. 9). In the absence of foreign threats, the expression of the *SpTrf* gene family is low or off in immunoquiescent sea urchins.^{14,28} Upon detection of foreign targets by the phagocytes, perhaps based on signaling from pathogen recognition receptors (PRRs) that are expressed by the cells,^{3,67–70} the *SpTrf* gene family is upregulated and the natSpTrf proteins are produced, loaded into transport vesicles, and secreted into the CF^{28,71} (Fig. 9A). Upon infection, monomeric natSpTrf proteins bind to the pathogens and undergo a conformational change from disordered to increased secondary structure and proceed to multimerization.^{27,31} This process results in increased affinity for the putative SpTrf-R on the surface of small and polygonal phagocytes, which leads to phagocytosis and clearance of the opsonized targets.³⁷ Signaling through the PRRs not only induces but maintains expression of the *SpTrf* gene family and perhaps other proinflammatory effectors such as *SpIL17-9*.

As pathogen clearance proceeds to eventual elimination from the CF, signaling from the PRRs decreases and eventually terminates. In the absence of pathogens, the relative concentration of soluble, unbound natSpTrf proteins in the CF become in excess (Fig. 9B). The monomers interact with each other, transform and multimerize, and increase their affinity for the SpTrf-R on phagocytes, resulting in their uptake and removal from the CF. Furthermore, the natSpTrf proteins bound to the SpTrf-R on phagocytes activates a putative signaling pathway. In the absence of PRR signaling, SpTrf-R signaling downregulates expression of at least some proinflammatory genes and the *SpTrf* gene family,³⁰ which leads to a reduction in the concentration of natSpTrf proteins in the CF. This working hypothesis for an immune response system in sea urchins includes controls for natSpTrf protein production and secretion when immune responsiveness is activated. It adjusts the concentration of the natSpTrf proteins in the CF for optimal function relative to the level of the foreign invasion, it clears unbound multimers from the CF, and it downregulates gene expression as the infection is resolved.³⁰ This process infers a putative auto-regulatory feedback signaling system to adjust immune gene expression relative to the level of the microbial invasion and clearance leading to the resolution phase of the immune response. This hypothesis for natSpTrf functions is consistent with how immune systems function and are regulated in all organisms.

A Pathogen invasion and immune response



B Pathogen clearance and immune resolution

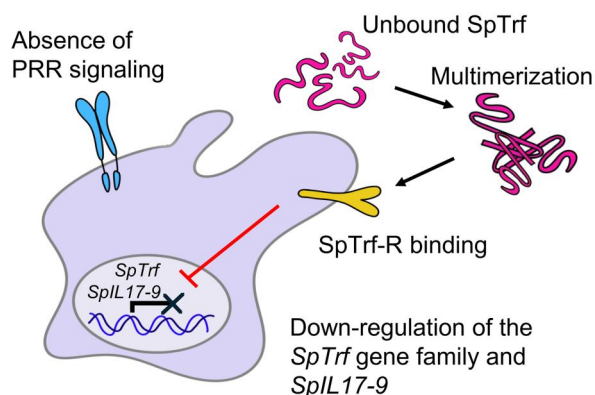


Figure 9. A working hypothesis for the functions of the natSpTrf system in sea urchins. The functions of the rSpTrf proteins isolated from *Escherichia coli* and insect cells illustrate a possible range of functions carried out by the natSpTrf proteins in sea urchins. (A) Pathogen detection likely occurs by PRRs (blue dimer) that signal immune activation and upregulate expression of the proinflammatory *SpIL17-9* gene and the *SpTrf* gene family leading to natSpTrf protein secretion. The natSpTrf proteins in the CF bind to pathogens, transform into disordered to increased secondary structure, and may multimerize. This increases their affinity for the putative SpTrf-Rs (yellow) displayed on phagocytes and induces phagocytosis of the pathogen. Signaling from the SpTrf-R is blocked or overridden by PRR signaling to continue *SpTrf* gene expression. (B) As the pathogens are cleared, the excess natSpTrf proteins in the CF, which multimerize, increase affinity for phagocytes and bind to putative cell surface receptors. In the absence of pathogen detection and PRR signaling, the cells respond to SpTrf-R signaling by downregulating expression of the *SpTrf* gene family and *SpIL17-9*. This decreases the natSpTrf proteins in the CF and eventually results in immune resolution.

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Supplementary material

Supplementary material is available at *The Journal of Immunology* online.

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Conflicts of interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Data availability

Data supporting the findings of this study are contained within the main paper and the supplementary material. Accession numbers for the proteins investigated in this study are included in Table S1.

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